

# Data Storm v7.0

- Preliminary Round -

## Summary Report:

Estimating Latent Demand via Spatial Signals and Peer Benchmarking

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## 1 1. Raw Data Ingestion (The Bronze Layer)

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Following standard Lakehouse architecture, our pipeline began with the Bronze Layer. We ingested all raw `.csv` files provided by the legacy Sales Force Automation (SFA) and distributor ERP systems exactly as-is. This immutable layer preserved the original state of the master data, spatial coordinates, and the two-year transaction history, ensuring we had a reliable, unmanipulated fallback before executing downstream data forensics.

## 2 2. Data Forensics and Hygiene (The Silver Layer)

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The primary engineering challenge transitioning to the Silver Layer was the sheer volume of the `transactions_history_final.csv` dataset. Standard Pandas processing caused memory exhaustion. To prevent session crashes and optimize processing time, we built our **DE checks setup** using **Polars**, streaming the raw CSV directly from disk.

Our pipeline trapped specific system anomalies to isolate true physical sales. Records failing these checks were **quarantined** from the final modeling layer:

- **Filtering Artificial Volume:** We trapped rows where `VolumeLiters == 0`. These represent system adjustments and operational fees rather than actual consumer demand and were quarantined from pricing math.
- **Deduplication:** We applied the `unique()` function to isolate and drop identical duplicate records generated by distributor handheld device upload failures.
- **Null Handling & Quarantine:** Any row missing a primary joining key (`Outlet_ID`) failed our referential integrity check and was strictly quarantined.
- **Geographic Preservation:** During the aggregation of historical volume, we specifically isolated and retained the `Distributor_ID` to dynamically map outlets back to their respective geographic provinces.

## 3 3. POI Data Acquisition (The Gold Layer)

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To estimate unconstrained potential, we needed an external spatial signal representing local foot traffic. We discarded sequential API loops (e.g., Google Places) due to rate limits and financial inefficiency, opting instead for a bulk-download approach.

### 3.1 Technical Approach & Data Sourcing

We utilized the **Overpass API via the OSMnx Python library** to download the complete spatial map data for the Western, Central, North-Western, and Southern provinces in a single execution. We specifically targeted the following **catchment drivers (POIs)**: schools, universities, colleges, hospitals, bus stations, and banks, as these are the primary generators of daily transient footfall.

### 3.2 Mapping to Internal Outlets

We used **GeoPandas** to map these external spatial signals to our internal `outlet_coordinates.csv`. Raw GPS coordinates in degrees distort distance calculations. We projected all spatial data to `EPSG:32644` (the accurate UTM mapping zone for Sri Lanka). This allowed us to calculate a mathematically precise 1,000-meter (1km) catchment radius around every internal outlet, performing a rapid spatial join (`sjoin`) to calculate the localized density of the targeted catchment drivers.

## 4 4. Causal Base Logic & Latent Potential Methodology

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The problem states that historical volume is left-censored—it represents only what was sold, limited by credit constraints, stockouts, or poor rep management. To calculate the uncapped ceiling, we deployed a **Peer Benchmarking** ("Look-Alike") strategy.

### 4.1 Mathematical Workflow

1. **Smoothing Holiday Seasonality:** The raw data contains extreme volume spikes during cultural holidays. To prevent the model from learning a false seasonal spike as a baseline, we calculated the aggregate volume per month and extracted the `Avg_Monthly_Volume`. This mathematically redistributed holiday peaks, giving us a stable baseline to predict the target month (January 2026).
2. **Hyper-Local Clustering:** We calculated a `Catchment_Score` for every outlet by summing their mapped POIs. We then grouped all shops into 10 distinct traffic deciles (`Traffic_Tier`). Finally, we combined the geographic province with the traffic tier (e.g., "Western\_9") to create hyper-local clusters based on environmental reality, completely ignoring outdated master-data labels like "Small" or "Large".
3. **Benchmarking the Star:** Within each granular cluster, we identified the 90th percentile performer. This "Star" outlet serves as our benchmark for a shop operating under perfect, unconstrained conditions (e.g., maximum fridge capacity, high credit limits).
4. **Transferring the Ceiling:** We uncapped the latent potential by setting the local Star's sales volume as the potential ceiling for every other shop in that exact micro-environment. If a shop was already outperforming the 90th percentile Star, we retained their actual `Avg_Monthly_Volume` using a `numpy.maximum` array evaluation.

### 4.2 Commercial Strategy

This mathematical framework shifts the business from rear-view allocation to predictive allocation. Instead of deploying coolers and marketing budgets to historically saturated outlets, the business can calculate the "Growth Gap." Investment should be heavily prioritized toward shops that sit in high-tier catchment environments (high POI density) but exhibit severely constrained historical sales.

## 5 5. GenAI Transparency Log

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During this 36-hour sprint, Generative AI was utilized strictly as an engineering accelerator for syntax and debugging, not as a replacement for analytical methodology.

- **Library Translation (Polars):** When Pandas encountered memory limits, we utilized AI to accurately translate our required DE forensic steps (deduplication, grouping, filtering) into Polars syntax for out-of-core execution.
- **Geospatial Accuracy:** We prompted the AI to identify the correct Coordinate Reference System (EPSG:32644) necessary for meter-based spatial buffering in Sri Lanka, bypassing manual documentation searches.
- **Dynamic Error Handling:** When standard column joins failed due to inconsistent legacy naming conventions in the Bronze data, AI was used to write a dynamic "fuzzy finder" script to locate and map the `Distributor_ID` field without breaking the pipeline.

- **Validation:** All AI-generated syntax was executed locally in our Jupyter environment and manually validated against the "4 Champions" methodology outlined in the competition brief.

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